

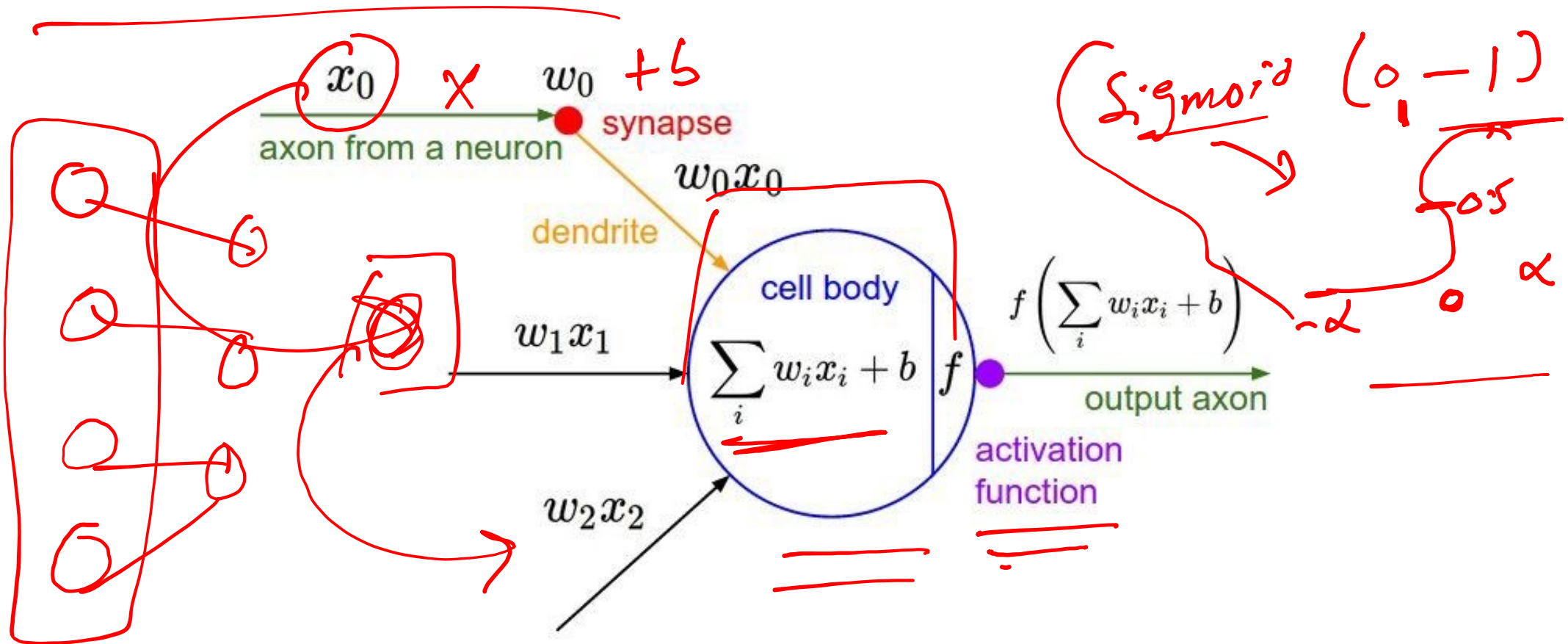
Activation Functions and Visualisation



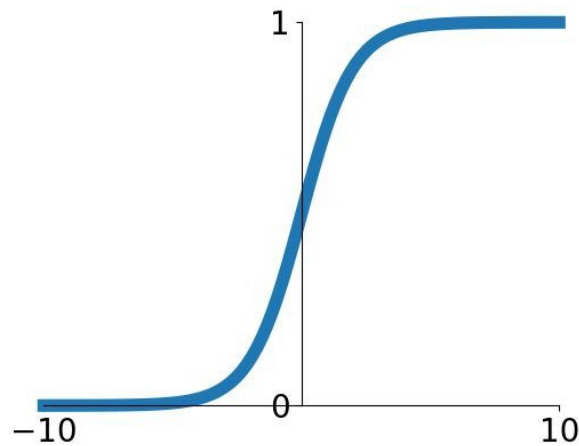
Activation Functions

Fei-Fei Li & Justin Johnson & Serena Yeung

Activation Functions



Activation Functions



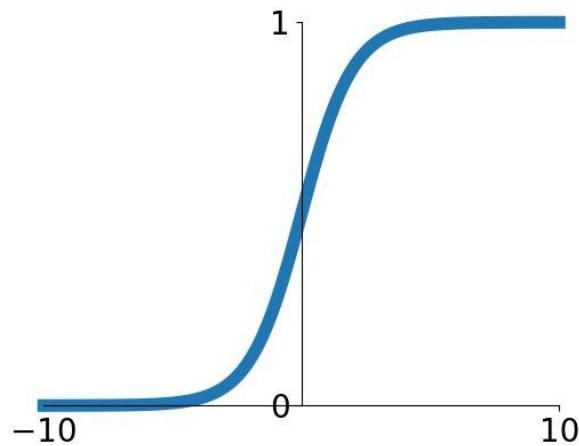
Sigmoid

output

$$\sigma(x) = 1 / (1 + e^{-x})$$

- Squashes numbers to range [0,1]
- Historically popular since they have nice interpretation as a saturating “firing rate” of a neuron

Activation Functions



Sigmoid

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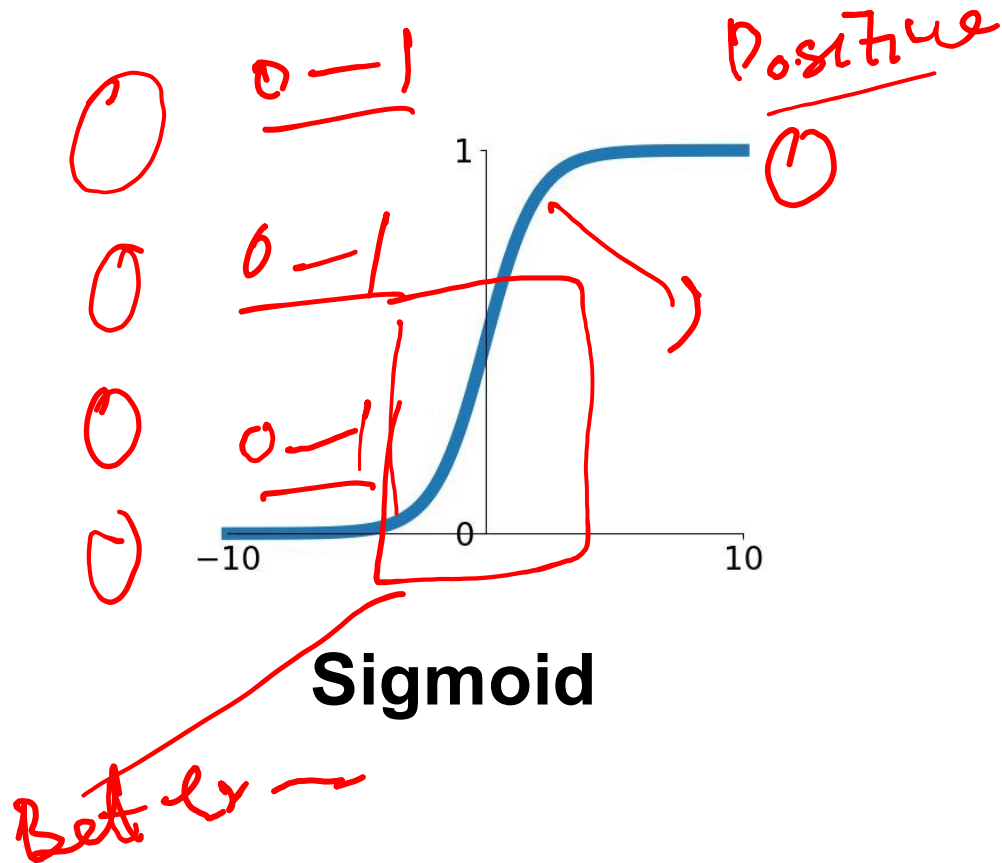
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3 problems:

1. Saturated neurons “kill” the gradients

Activation Functions

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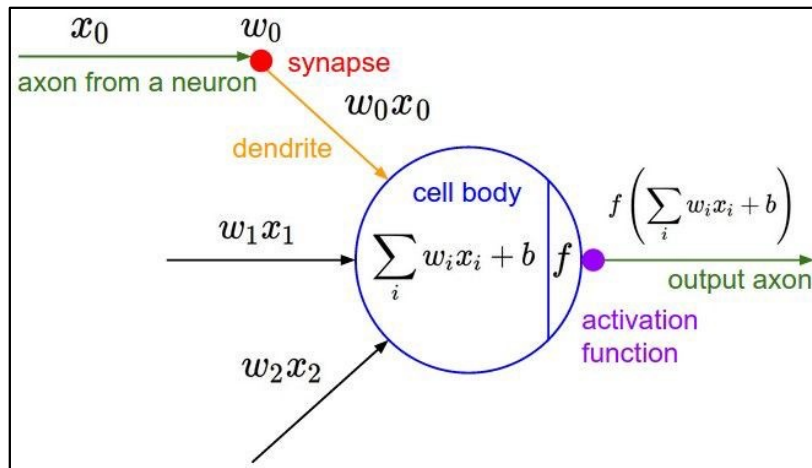


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3 problems:

1. Saturated neurons "kill" the gradients
2. Sigmoid outputs are not zero-centered

Consider what happens when the input to a neuron (x) is always positive:



$$f\left(\sum_i w_i x_i + b\right)$$

What can we say about the gradients on $\mathbf{w}$?

Consider what happens when the input to a neuron is always positive...

$$f\left(\sum_i w_i x_i + b\right)$$

allowed
gradient
update
directions

allowed
gradient
update
directions

zig zag path

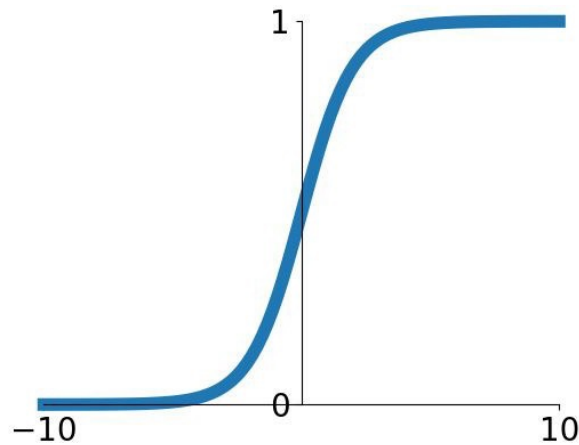
hypothetical
optimal w
vector

What can we say about the gradients on $\underline{w}$?

Always all positive or all negative :(

(this is also why you want zero-mean data!)

Activation Functions



Sigmoid

$$\sigma(x) = 1/(1 + e^{-x})$$

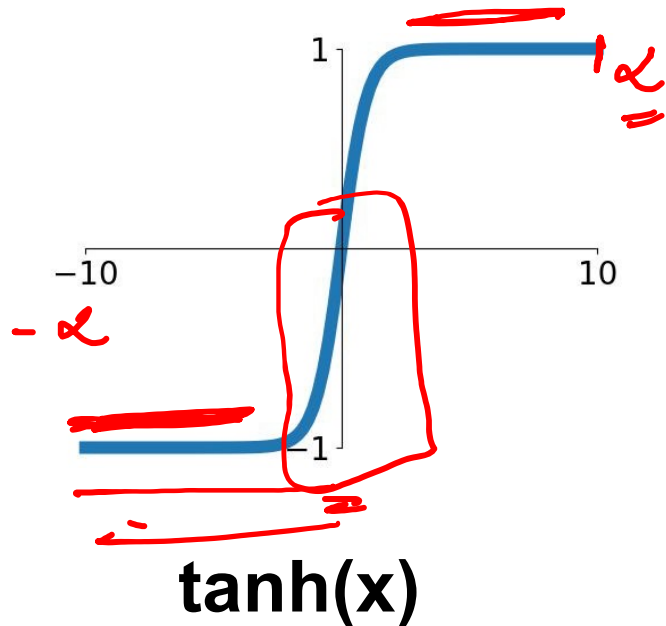
- Squashes numbers to range [0,1]
- Historically popular since they have nice interpretation as a saturating “firing rate” of a neuron

3 problems:

Handwritten note: e^{-x} as $x \rightarrow \infty$

1. Saturated neurons “kill” the gradients
2. Sigmoid outputs are not zero-centered
3. $\exp()$ is a bit compute expensive

Activation Functions

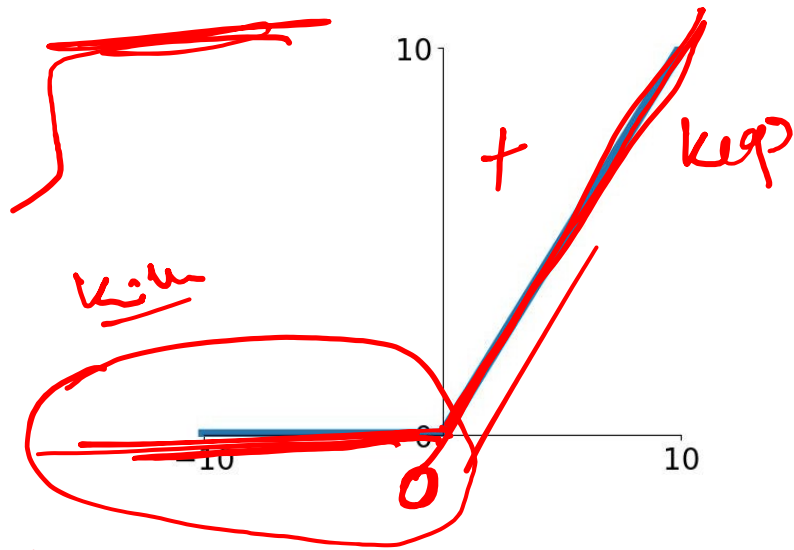


$$\frac{e^x - e^{-x}}{e^x + e^{-x}}$$

- Squashes numbers to range $[-1,1]$
- zero centered (nice)
- still kills gradients when saturated :(

[LeCun et al., 1991]

Activation Functions



- Computes $f(x) = \max(0, x)$

- Does not saturate (in +region)

- Very computationally efficient

→ - Converges much faster than sigmoid/tanh in practice (e.g. 6x)

- Actually more biologically plausible than sigmoid

ReLU

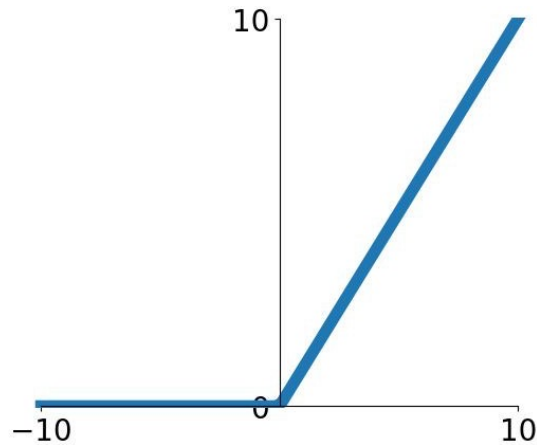
(Rectified Linear Unit)

$$\max(\underline{u}, 0)$$

e^x

[Krizhevsky et al., 2012]

Activation Functions

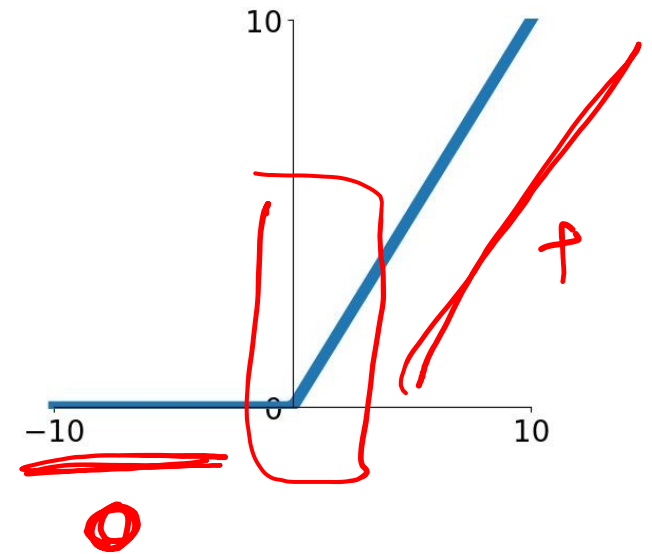
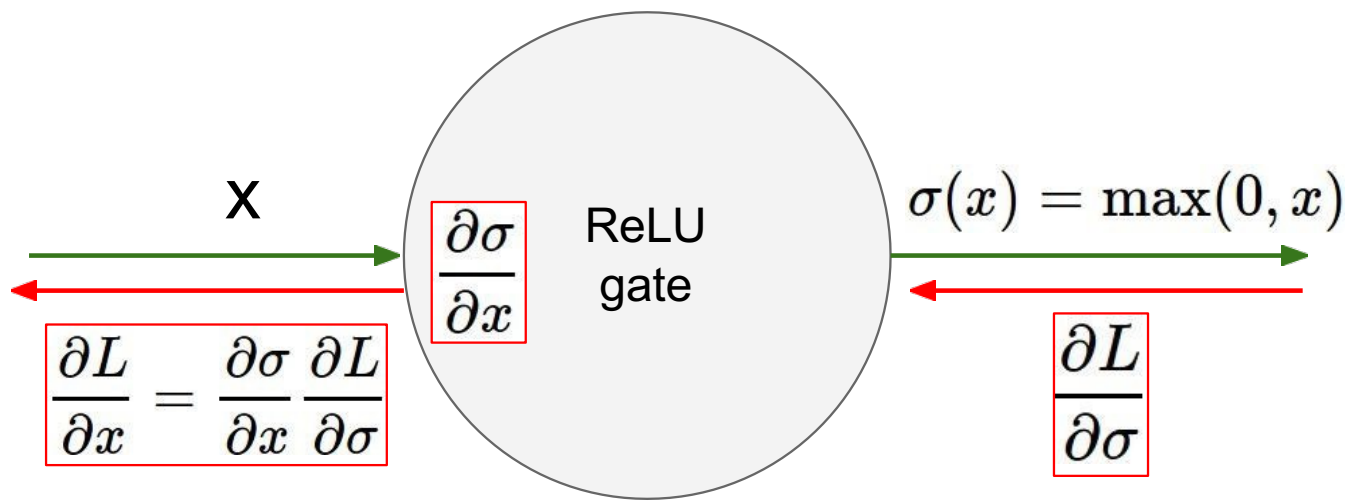


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ReLU
(Rectified Linear Unit)

- ⇒ - Not zero-centered output
- An annoyance:

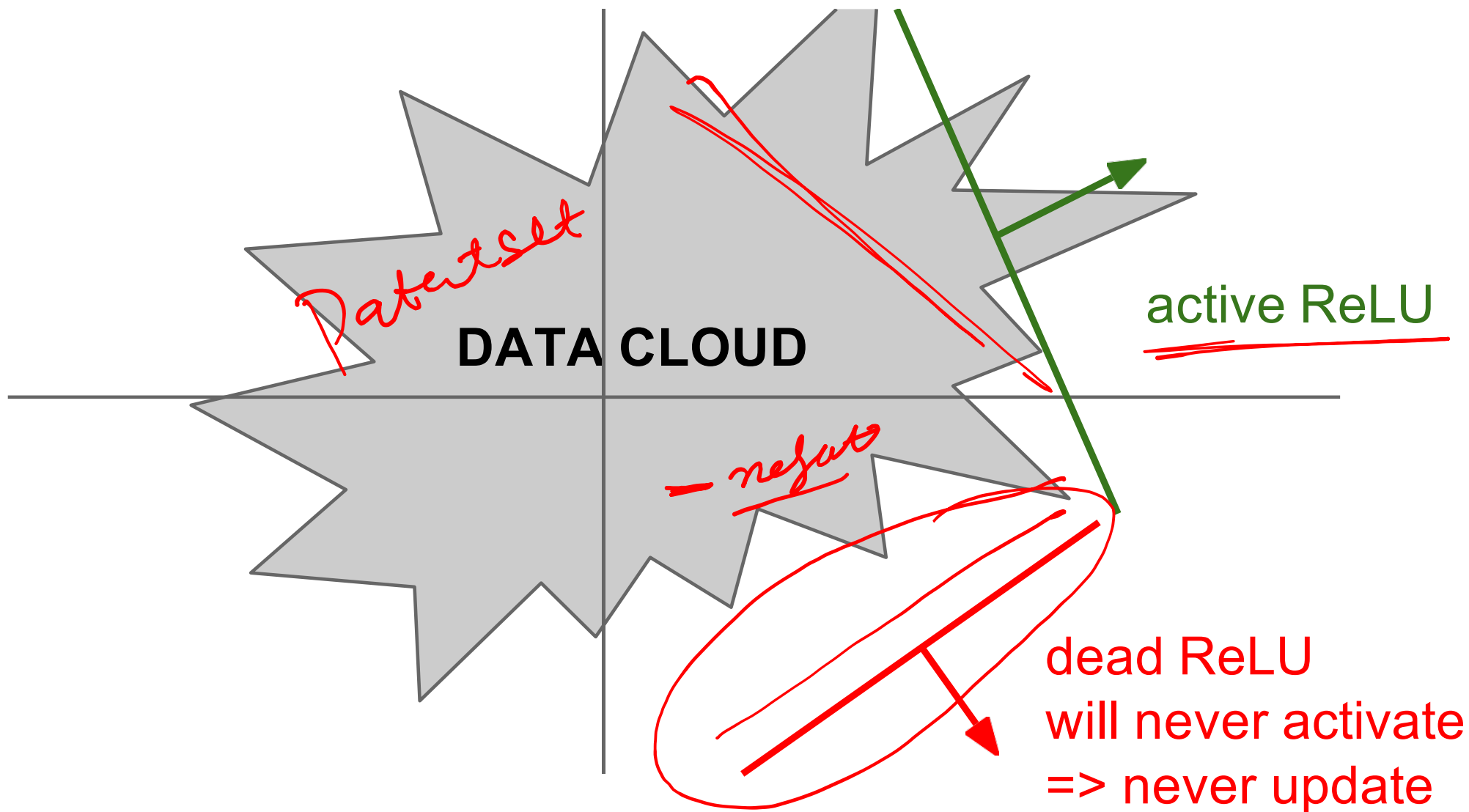
hint: what is the gradient when $x < 0$?

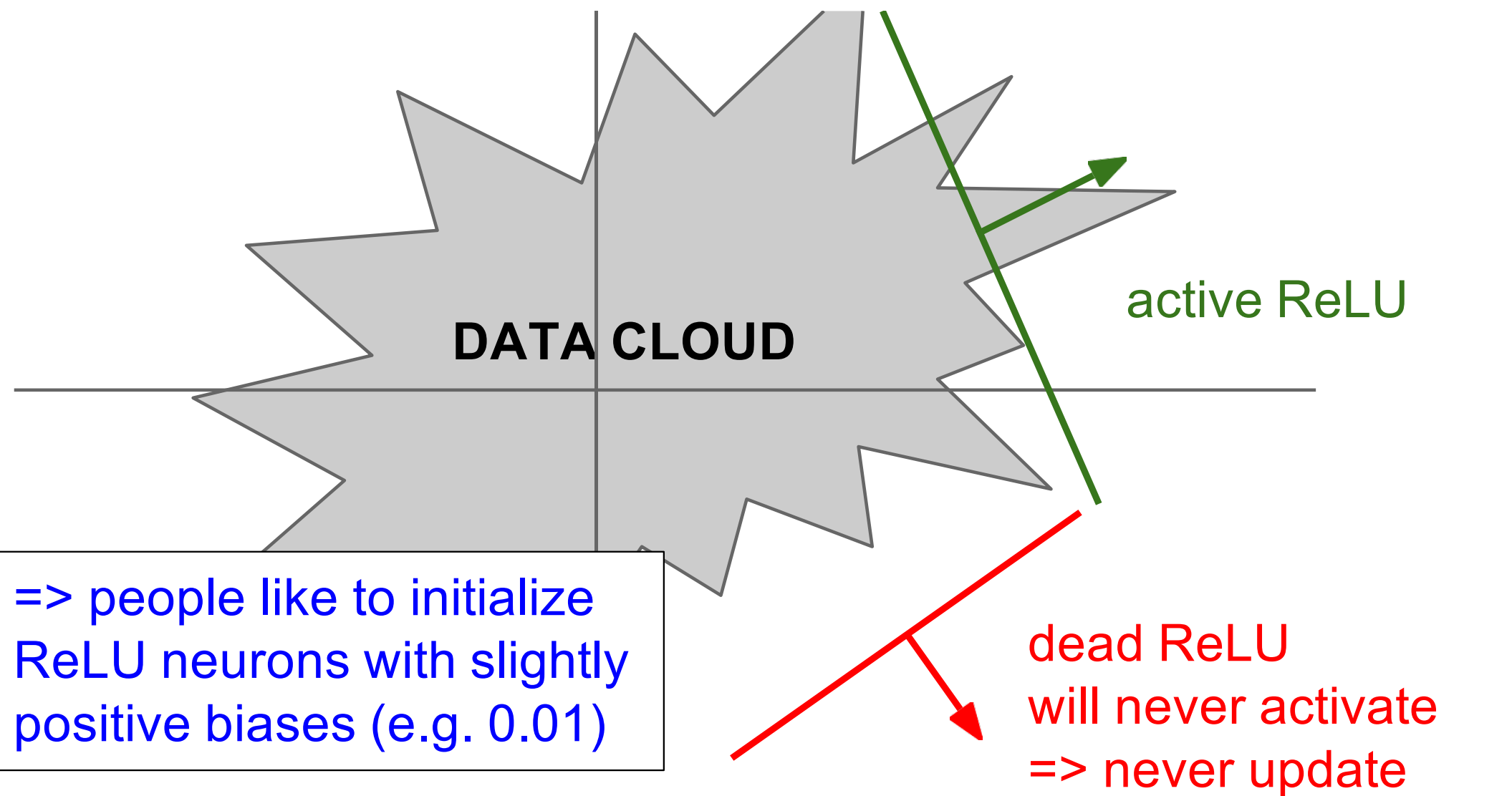


What happens when $x = -10$?

What happens when $x = 0$?

What happens when $x = 10$?





A diagram illustrating the concept of a 'DATA CLOUD' and its relationship to ReLU neurons. A gray, star-shaped region represents the 'DATA CLOUD'. A horizontal line and a vertical line intersect at the center of the cloud. A green line with an arrow pointing upwards and to the right is labeled 'active ReLU'. A red line with an arrow pointing downwards and to the right is labeled 'dead ReLU will never activate => never update'.

DATA CLOUD

active ReLU

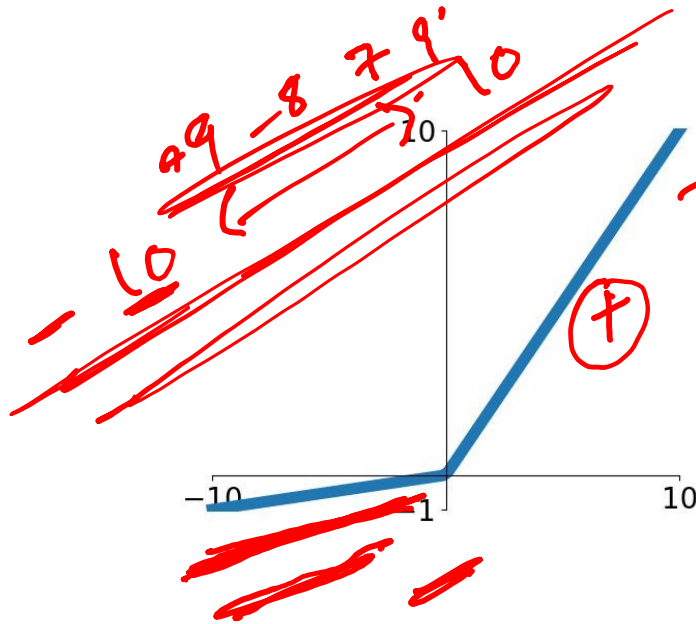
=> people like to initialize
ReLU neurons with slightly
positive biases (e.g. 0.01)

dead ReLU
will never activate
=> never update

Activation Functions

[Mass et al., 2013]

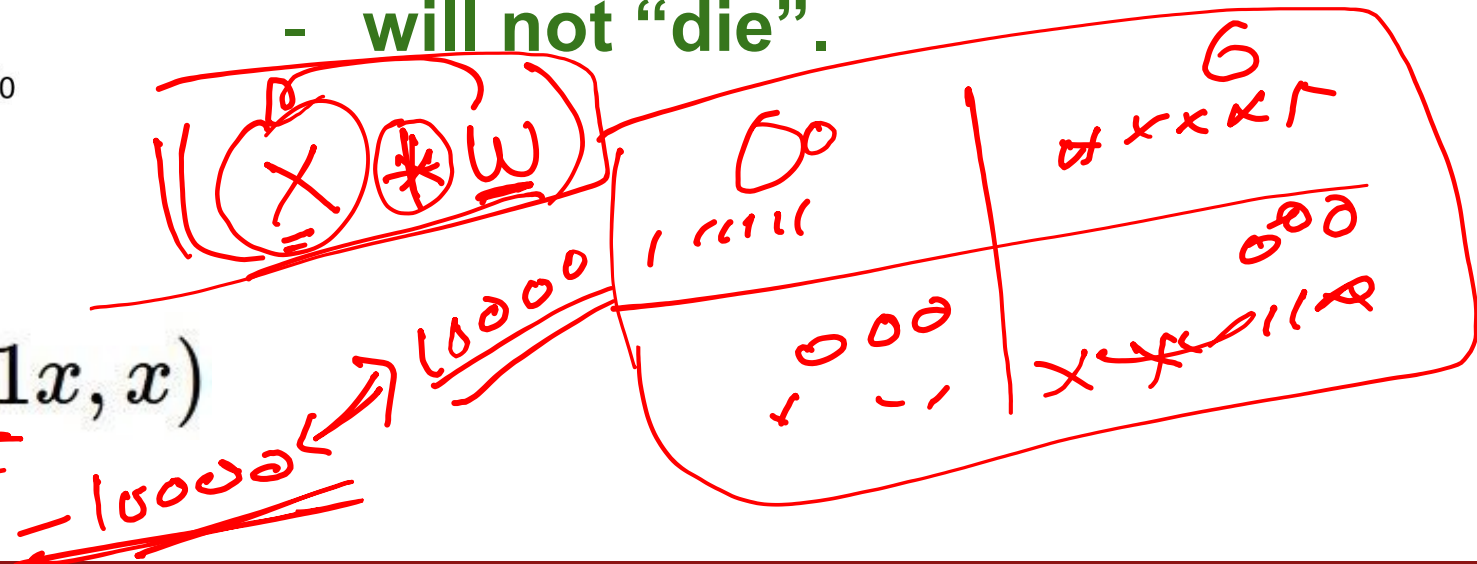
[He et al., 2015]



Leaky ReLU

$$f(x) = \max(0.01x, x)$$

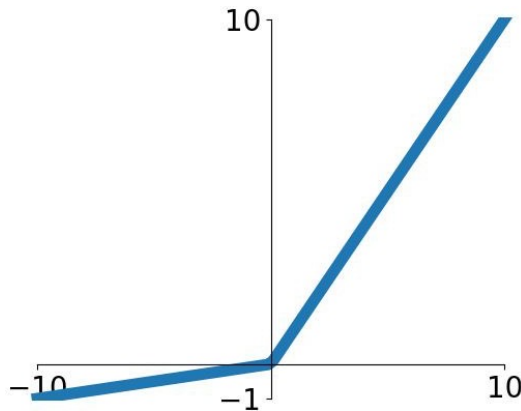
- Does not saturate
- Computationally efficient
- Converges much faster than sigmoid/tanh in practice! (e.g. 6x)
- **will not “die”.**



Activation Functions

[Mass et al., 2013]

[He et al., 2015]



Leaky ReLU

$$f(x) = \max(\underline{0.01x}, x)$$

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- **will not “die”.**

Parametric Rectifier (PReLU)

$$f(x) = \max(\underline{\alpha x}, x)$$

backprop into α
(parameter)

Visualisation Libraries



